Discrete Counting by Floors, Binomials, Incidences, and Symmetries

N-20221209

A set-builder expression is an instruction to select objects. A binomial coefficient counts choices of positions; an incidence table can be counted by rows or columns; a group action identifies objects that a symmetry makes equivalent. The natural order for this exercise sheet is to turn every symbolic condition into a concrete count. The standard techniques follow Richard Stanley's *Enumerative Combinatorics, Volume 1* and the MIT 18.212 combinatorics notes.[1]

§1.1 Set-builder notation forces us to fix the universe

The expression

$$S = \{x \in \mathbb{R} : 3\lfloor x \rfloor - 5 < 0\}$$

is read by solving the condition $3\lfloor x \rfloor < 5$. Since $\lfloor x \rfloor$ is integral, this is equivalent to $\lfloor x \rfloor \leq 1$, hence

$$S = (-\infty, 2).$$

The endpoint 2 is excluded: $\lfloor 2 \rfloor = 2$ violates the inequality. If the universe had been $\mathbb{Z}$, the answer would instead be the integers at most 1. The universe is part of the problem, not an irrelevant heading.

Set notation also distinguishes $a \in A$ from $\{a\} \subseteq A$. The first is a membership statement about an element; the second is a containment statement about a set. When $a \in A$, both can be true, but they have different types and behave differently under operations on sets.

§1.2 A floor sum counts lattice points

For integers $a, m > 0$, the sum

$$S(a, m) = \sum_{k=0}^{m-1} \left\lfloor \frac{ak}{m} \right\rfloor$$

counts the integer lattice points

$$\{(k, j) : 0 \leq k < m, \quad 1 \leq j \leq ak/m\}.$$

A useful exact formula arises when $\gcd(a, m) = 1$:

$$\boxed{\sum_{k=1}^{m-1} \left\lfloor \frac{ak}{m} \right\rfloor = \frac{(a-1)(m-1)}{2}.}$$

[1] MIT 18.212, *Algebraic Combinatorics*, lecture notes.

For $1 \le k \le m-1$, the rational number ak/m is not integral. Pair the terms indexed by k and $m-k$:

$$\left\lfloor \frac{ak}{m} \right\rfloor + \left\lfloor \frac{a(m-k)}{m} \right\rfloor = a - 1.$$

Summing this identity over $k = 1, \dots, m-1$, each floor term is counted twice. The displayed formula follows. The coprimality hypothesis matters; it ensures there are no lattice points on the diagonal at interior integer coordinates.

For $a = 3, m = 5$, the floors are $0, 0, 1, 1, 2$, summing to 4, and the formula gives $(3-1)(5-1)/2 = 4$. Here a floor function became a lattice-point problem with a symmetry proof.

§1.3 Binomial identities count nested choices

The identity

$$\sum_{j=0}^{n} \binom{n}{j} = 2^n$$

counts all subsets of an n-element set by partitioning them according to cardinality. Its signed companion

$$\sum_{j=0}^{n} (-1)^j \binom{n}{j} = 0, \qquad n \ge 1,$$

follows by pairing each subset not containing a fixed element with its version containing that element. The two subsets have opposite parity.

Another identity,

$$\binom{n}{r}\binom{n-r}{k-r} = \binom{n}{k}\binom{k}{r} \qquad (0 \le r \le k \le n),$$

counts ordered pairs $R \subseteq K \subseteq [n]$ with $|R| = r$ and $|K| = k$. The left side chooses R first, then extends it to K; the right side chooses K first, then selects R inside it. The equality is a double count with the same combinatorial objects on each side.

§1.4 Finite Differences and Binomials

Define $\Delta f(n) = f(n+1) - f(n)$. Pascal's identity gives

$$\Delta \binom{n}{k} = \binom{n}{k-1}.$$

For a polynomial f of degree at most d, its forward differences satisfy $\Delta^{d+1} f = 0$. Successive application at $n = 0$ yields the Newton expansion

$$\boxed{f(n) = \sum_{k=0}^{d} (\Delta^k f)(0) \binom{n}{k}}$$

for nonnegative integers n, with the polynomial identity extending it to all n.

For $f(n) = n^2$, the initial differences are $f(0) = 0$, $\Delta f(0) = 1$, $\Delta^2 f(0) = 2$, so

$$n^2 = \binom{n}{1} + 2\binom{n}{2} = n + n(n-1).$$

This is why the binomial basis naturally belongs to discrete counting: its differences behave like the derivatives of ordinary powers but with no denominators.

§1.5 Recursive sets and finite stabilization

Suppose a rule $F : \mathcal{P}(U) \to \mathcal{P}(U)$ is extensive, so $S \subseteq F(S)$, and U is a finite set of size N. Starting from S_0 and setting $S_{k+1} = F(S_k)$ gives an increasing chain

$$S_0 \subseteq S_1 \subseteq S_2 \subseteq \cdots .$$

Every strict step adds at least one new element. After at most $N - |S_0|$ strict steps the process stabilizes at $S_* = F(S_*)$. The bound does not require that F be monotone on all subsets; extensivity is enough for this particular increasing chain.

If a recursive rule is not extensive, the sets may cycle. For $U = \{1\}$, the operation $F(S) = U \setminus S$ alternates between $\varnothing$ and U forever. Hence finiteness of the universe alone does not imply stabilization of an arbitrary recurrence.

§1.6 Incidence counting is a matrix identity

Let X be a finite set of elements and $\mathcal{A}$ a finite family of subsets of X. Define the incidence set

$$I = \{(x, A) \in X \times \mathcal{A} : x \in A\}.$$

Counting first by x, then by A, gives

$$\boxed{\sum_{x \in X} \#\{A \in \mathcal{A} : x \in A\} = \sum_{A \in \mathcal{A}} |A|.}$$

For example, if $X = \{1, 2, 3, 4\}$ and $\mathcal{A}$ consists of all two-element subsets, there are $\binom{4}{2} = 6$ members, so the right side is $6 \cdot 2 = 12$. On the left, each point belongs to $\binom{3}{1} = 3$ two-element subsets, giving $4 \cdot 3 = 12$. In matrix language, both sides sum all entries of the same 0-1 incidence matrix, in different orders.

§1.7 Symmetry counting needs stabilizers

Suppose a finite group G acts on a finite set X. Simply dividing $|X|$ by $|G|$ assumes every orbit has the same size $|G|$. In reality,

$$|Gx| = \frac{|G|}{|\operatorname{Stab}(x)|}.$$

A correct general formula is Burnside's lemma:

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|.$$

For binary necklaces of length three, $G = C_3$ acts on the $2^3 = 8$ binary strings by cyclic rotation. The identity fixes all 8. Each nontrivial rotation fixes only 000 and 111, hence exactly 2 strings. Therefore the number of rotation classes is

$$\frac{8 + 2 + 2}{3} = 4.$$

The naive quotient $8/3$ was not even an integer. Fixed-point counting repairs it because not every string has an orbit of size three.

§1.8 The main diagnostic for a counting problem

The correct move depends on what is being counted. Floors encode lattice-point thresholds; binomial coefficients encode nested finite choices; finite differences encode discrete polynomial changes; recursive sets demand explicit stability conditions; incidence arguments count a single relation in two ways; symmetry counts require orbit and stabilizer information. A useful explanation always identifies the actual objects and proves a formula for them. Additional connections to Ehrhart theory, polynomial interpolation or graph spectra are worthwhile only when they lead to a new mathematical computation.